

(54) METHOD AND APPARATUS FOR ADAPTIVE QUERY ROUTING

(57) Abstract

A method for adaptive routing of database queries across heterogeneous backends using a learned cost model. The system selects a backend based on predicted latency, freshness, and load, and adapts the model online from execution feedback.

Claims

1. A method comprising: receiving a query Q at a router; computing a feature vector F from Q; predicting, via a cost model M, an expected latency for each backend in a set B; and dispatching Q to $\text{argmin}_b \text{latency}(b)$.
2. The method of claim 1, wherein computing F includes hashing the query plan and concatenating cardinality estimates.
3. The method of claim 1, wherein M is a gradient-boosted regressor updated nightly from execution traces.
4. The method of claim 3, wherein updating M comprises minimising a Huber loss over (F, observed_latency) tuples.
5. The method of any of claims 1 to 4, further comprising falling back to a default backend when predicted latency exceeds a threshold T.
6. An apparatus comprising memory and one or more processors configured to perform the method of any one of claims 1 to 5.
7. A non-transitory computer-readable medium storing instructions that, when executed, cause performance of the method of claim 1.